



**FIRST SEMESTER 2026-27
COURSE HANDOUT**

Date: 01.08.2026

In addition to part I (general handout for all courses appended to the Time Table), this portion gives further specific details regarding the course.

Course No. : **PHA F 311**
Course Title : **Pharmacology-I**
Instructor-In-Charge : **Prof. Rajeep Taliyan**

Team of Instructors : Mr. Jayant Singh Bhardwaj, Talati Yugant Krishnakumar, Bhadange Rohan Satish, Hemraj Singh, Saral Bachhuka

1. Course Description:

Pharmacology of important classes of drugs including their mechanisms of action, therapeutic uses, side effects, toxic manifestations, indications and contraindications.

2. Scope and Objectives:

This course is intended to impart the knowledge regarding the sources, routes of drug administration, pharmacokinetics (ADME) and pharmacodynamics (mechanism of action) of various drugs. This course is also intended to impart the knowledge regarding the concepts of action of drugs on various systems of the human body including ANS, SNS and CNS. The course also deals with therapeutic uses, side effects and contraindications of the drugs, which are commonly prescribed for the treatment of various disease conditions.

3. Text book (TB):

1. Tripathi KD, Essentials of Medical Pharmacology, 8th edition, Jaypee brothers, New Delhi, 2022
2. Satoskar R.S. & Bhandarkar S.D., Pharmacology and Pharmacotherapeutics 25th edition, Popular Prakashan, Bombay, 2017

4. Reference Books (RBs):

1. Lippincott's Illustrated reviews: Pharmacology, Karen Whalen, MJ Mycek et al (Editor), Lippincott-Raven Publishers, New York, 7th edition, 2018
2. Katzung BG, Basic and Clinical Pharmacology, 16th edition, McGraw-Hill, 2023
3. Goodman & Gilman's The Pharmacological Basis of Therapeutics, by Laurence L. Brunton (Editor), 14th edition, McGraw-Hill Publishing Co, 2022.
4. Pharmacology, HP Rang, MM Dale & JM Ritter (editors), 6th edition, Churchill Livingstone, 2007.





5: COURSE PLAN:

A. Theory Components

Module Number	Learning Objectives	Reference (TB) /RB	Learning Outcome
1: Introduction: General Pharmacology	L1: 1. Introduction to Pharmacology, Scope, Sources of drugs, Dosage forms	Ch-T-1: R-3 (1)	Understand pharmacology, its scope, Sources of drugs and dosage forms,
	L1: 2. Routes of drug administration	R-3: Ch (3)	Different route of drug administration and their influences /modulation on the physiological process.
	L1:3-6. Pharmacokinetics, Pharmacodynamics: mechanism of action,		Understanding, what drug does to the body or what body does to drugs (ADME)?
	L1:7-8. Combined effect of drugs, Factors modifying Drug action,	Ch-T-I	Study and understand the effect of combination of two or more drugs
	L1:9.0 Drug tolerance, dependence and drug addiction		Study and understand the process of Drug addiction, Tolerance, Drug dependence
2. Introduction to Cardiovascular agents	L2: 1.4: Basic mechanisms of heart functions and disorders, ECG interpretation, Myocardial Infarction		Understand different classes of drugs used for the management of CVS disorders, MOA and their side effects
3: Introduction to Inflammation and pain mediators, Autocoids synthesis and release in body,	L3 1-3 Classification, Pharmacology of NSAIDS and centrally acting agents	Ch-T-V	Understand the process of pain and different classes of drugs, MOA, side effects of NSAIDS, centrally acting analgesics: Opioid analgesics





	analgesics		and toxicity
4. Introduction to Nervous system,	L4:1-2. Classification and Nervous system, general actions	Ch-T-II	General Introduction to nervous system, Neurohumoral transmission (ANS) Parasympathomimetic (Cholino-mimetic drugs), Anti-cholinergic and anti-adrenergic agents (alpha and beta blockers) Mydriatics and Cycloplegics Muscle relaxant Management of Myasthenia Gravis and glaucoma
5: Drugs acting on ANS	L4:3-6. Pharmacology: Drugs acting on ANS	Ch-T-IV, R-	
6. Introduction to Central Nervous system (CNS)	L6: 1-3. Introduction to CNS and Drugs acting on CNS	Ch-T-IV	Understand different classes of drugs, MOA, side effects of drugs used for the management of CNS disorder/disease
	L6:4-10. Pharmacology of CNS Disorders and their management	CH-R-II	
7: Introduction to Adverse drug reactions	L7: 1-2. Adverse drug reactions, Pharmacogenetics, iatrogenic reaction and their monitoring,	Ch-T-I	Study and understand the mild, moderate and severe -life threatening adverse reaction of drugs and their management
8: Introduction of allergic and autoimmune disease	L8:1-3. Classification- uses of anti-histaminic agents, Anti-asthmatics, Auto-immune disease: Type-1-DM. RA	Ch-T-V II	Understanding the formation and physiological and pathological effects of autacoids, Classes of drugs, MOA and pharmacological uses of drugs that modulates autacoids formation/actions.



B. Practical components:

S.No	Experiments	Schedule
10-15	Experiments using common laboratory animals and software based, related to CVS, CNS/ ANS, pain and inflammation	10-15-Experiments for each laboratory section

6: Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close/Open -Book)
Mid-Semester Test	90 Min.	28	07/10/26	CB
Comprehensive Examination	180 Min.	40	10/12/26	OB/CB
Continuous assessment	-	32	-	CB /OB

*Continuous assessments will be based on the theory covered, Laboratory work, home assignments/ Quiz /case study, class presentations and Viva-Voce

Attendance: Although attendance is not compulsory, regularity in theory and practical classes will be a decisive factor during grading, especially in borderline cases. Laboratory attendance is a must and no make-up will be given for it (except medical urgency).

7: Chamber Consultation Hour: Saturday 12.00- 1.0 pm

8: Notices: Concerning this course will be displayed on Department of Pharmacy notice board/Nalanda or via email.

9: Make-up policy: Generally, make-up will be considered for regular students only as per the university rule. It is solely dependent on the “genuineness” of the circumstances. The make-up application should be personally given to instructor-in-charge.

Instructor-in-charge

PHA F 311

